

GARTH DUSTIN P. AYANG-ANG

FULL-STACK SOFTWARE ENGINEER | BACKEND SYSTEMS & BUSINESS AUTOMATION

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PROFESSIONAL SUMMARY

Full-stack software engineer with 3+ years building production business systems, currently focused on backend architecture, third-party API integration, and business process automation for a multi-tenant SaaS platform. Converts manual, staff-driven workflows into durable automated pipelines — self-service customer onboarding, online payment collection, and booking-to-payment flows — using TypeScript, Node.js, PostgreSQL, event-driven queues, and REST/webhook integrations across Stripe, telephony, and AWS. Works the full ownership cycle on a system: assess the operational bottleneck, choose the architecture, build and integrate it, automate it, document it, and monitor it in production.

CORE COMPETENCIES

Systems Architecture • Business Process Automation • REST API Design & Integration • Webhooks & Event-Driven Workflows • Online Payment Processing • Booking & Reservation Systems • Multi-Tenant SaaS • Relational Database Design • Third-Party API Research & Onboarding • Technical Documentation • Requirements Gathering with Non-Technical Stakeholders

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C#, Python, SQL, Kotlin

Backend & APIs: Node.js, Hono, REST API design, Webhooks, JSON/XML payload parsing, Zod validation, OpenAPI/Swagger, JWT & API-key authentication, ASP.NET Core, .NET

Frontend: Svelte 5 / SvelteKit, HTML, CSS/SASS, Tailwind CSS, shadcn, Angular, jQuery, Bootstrap, responsive design

Databases: PostgreSQL (Neon, Supabase), Microsoft SQL Server, Drizzle ORM, Entity Framework, schema design, migrations, stored procedures

Automation & Orchestration: Inngest durable workflows, BullMQ, Redis queues, saga / compensating-transaction patterns, scheduled jobs, idempotent retry design

Integrations: Stripe (Connect, Checkout, webhooks, metered billing), Vonage (Voice & Messages APIs, subaccounts, SIP), Twilio, LiveKit, Dentally practice-management API, SMS & transactional email automation

Cloud & DevOps: AWS (S3, Secrets Manager, KMS), Git & GitHub, Turborepo / pnpm monorepos, Vitest, CI workflows, Vercel / Node deployments

AI & LLM Tooling: OpenAI API, LangChain (supporting work alongside a dedicated AI team)

PROFESSIONAL EXPERIENCE

Dentalflo AI — Remote (Australia)

August 2025 – Present

Full-Stack Software Engineer

Backend-focused engineer on a multi-tenant dental practice-management and patient-communication platform (TypeScript, Hono, SvelteKit, PostgreSQL, Redis, BullMQ, Inngest). Personally own tenant provisioning, telephony integration, and online payments.

- Architected and shipped the platform's automated tenant provisioning pipeline — a durable, idempotent saga spanning five external systems (Neon PostgreSQL, AWS Secrets Manager, authentication, schema migrations, transactional email) that converts a paid signup into a fully configured, live clinic workspace with zero manual setup by internal staff.
- Designed a Redis-backed compensation (rollback) stack so any failed provisioning step automatically unwinds every step already completed, eliminating half-created tenants and the manual cleanup they previously required; behavior is covered by unit tests.
- Re-platformed provisioning from a seven-job BullMQ flow onto Inngest durable execution, giving each step independent retries and a single declarative failure handler — removing seven copies of duplicated error handling and making signup resilient to Stripe webhook retries.
- Implemented the Vonage telephony integration end to end: a subaccount-binding saga (application creation, phone-number linking, API user and credential provisioning, secrets storage, tenant-database writes) with full compensating actions, plus inbound voice, SMS, and delivery-status webhooks that bridge PSTN calls into LiveKit SIP.
- Migrated the platform from the legacy Vonage Voice API to the Messages API across all tenants, including a per-tenant capability declaration and a backfill script, retiring the deprecated webhook configuration without customer downtime.

- Built the Stripe Connect booking-deposit system, allowing patients to pay a deposit directly from an SMS or phone booking: connected-account onboarding with country and currency handling, Checkout sessions with enforced expiry windows, branded short payment links, signature-verified webhooks, and automatic write-back of payment status and appointment state to the clinic's practice-management system — removing manual invoicing and deposit chase-up calls from front-desk staff.
- Hardened the payment flow against duplicate and orphaned charges: enforced one payment link per appointment (unique index plus de-duplication migration), auto-voided pending links once a patient attends, synchronized deposit prices from Stripe, and automatically disabled deposit collection when a clinic disconnects its Stripe account.
- Rebuilt inbound SMS handling on a Redis queue and Inngest pipeline with typed delivery metadata, surfacing per-message delivery state directly in the staff conversation view and replacing a fragile single-worker design.
- Built the internal admin portal in SvelteKit — real-time provisioning and deprovisioning progress over server-sent events, tenant lifecycle controls, phone-number binding, and self-service password reset across two applications — reducing engineer involvement in day-to-day account operations.
- Shipped the public marketing site in SvelteKit, including page structure, SEO metadata, pricing pages, and performance/loading behavior, connecting the customer-facing front end to the same backend that provisions accounts.

Tropical Focus Philippines Inc. — Pasig City, Philippines

September 2024 – August 2025

Software Engineer / Business Analyst

- Integrated enterprise building-automation software with IoT devices and sensors, collecting real-time telemetry into Microsoft SQL Server to drive energy-cost optimization across large-scale commercial buildings.
- Served as both developer and business analyst: gathered requirements directly from mechanical engineers, energy consultants, and project managers, then translated operational problems into technical specifications and shipped them in C# / ASP.NET Core.
- Delivered data-access and reporting layers with stored procedures and Entity Framework, and worked cross-functionally to tailor deployments to each client site.

ScaleUp Solutions, Inc. — Quezon City, Philippines

June 2023 – December 2023

Web Developer (Part-Time) / Web Development Intern

- Integrated ERP systems with e-commerce platforms so product, order, and inventory data flowed between them automatically, eliminating duplicate manual entry for client staff.
- Built responsive, mobile-friendly storefront interfaces with HTML, SASS, JavaScript, ASP.NET Core, jQuery, and Bootstrap, applying modern UX practices to improve usability.
- Worked directly with clients to capture business requirements and report project progress.

EDUCATION

University of Saint Louis Tuguegarao — Tuguegarao City, Philippines

2020 – 2024

Bachelor of Science in Computer Science — Cum Laude

- Thesis: multi-label retinal disease classification on a high-class-imbalanced fundus image dataset (Python).
- Selected projects: university events management system (ASP.NET Core); online voting system (C#); Gelic, a dynamically typed C-syntax programming language with arithmetic, conditionals, and loops, written in Python 3.

ACHIEVEMENTS & PUBLICATIONS

- AI research paper accepted at ICITE 2023 (International Conference on Information Technology and Education), Boracay, Philippines — November 2023.
- Ranked 2nd, DICT Philippine Startup Challenge 7 Regional Pitching Competition — October 2022.